EXAMPLE 5.2

The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium.

Solution

Dielectric constant (relative permittivity of the medium), $\epsilon_r = 2.25$

Magnetic permeability, $\mu_r = 2.5$

Refractive index of the medium,

$$n = \sqrt{\epsilon_r \mu_r} = \sqrt{2.25 \times 2.5} = 2.37$$

5.2.2 Sources of electromagnetic waves

Any stationary charge produces only electric field (refer Unit 1). When the charge moves with uniform velocity, it produces steady current which gives rise to magnetic field (not time dependent, only space dependent) around the conductor in which charge flows. If the charged particle accelerates, it produces magnetic field in addition to electric field. Both electric and magnetic fields are time varying fields. Since the electromagnetic waves are transverse waves, the direction of propagation of electromagnetic waves is perpendicular to the planes containing electric and magnetic field vectors.

Any oscillatory motion is also an accelerated motion. So, when the charge oscillates (oscillating molecular dipole) about their mean position (Figure 5.9), it produces electromagnetic waves.

Suppose the electromagnetic field in free space propagates along z-direction and if the electric field vector points along x-axis, then the magnetic field vector will be mutually perpendicular to both electric field and the direction of wave propagation. Thus

$$E_x = E_\circ \sin(kz - \omega t)$$

$$B_y = B_\circ \sin(kz - \omega t)$$

where $E_\circ$ and $B_\circ$ are amplitudes of oscillating electric and magnetic field, k is a wave number, ω is the angular frequency of the

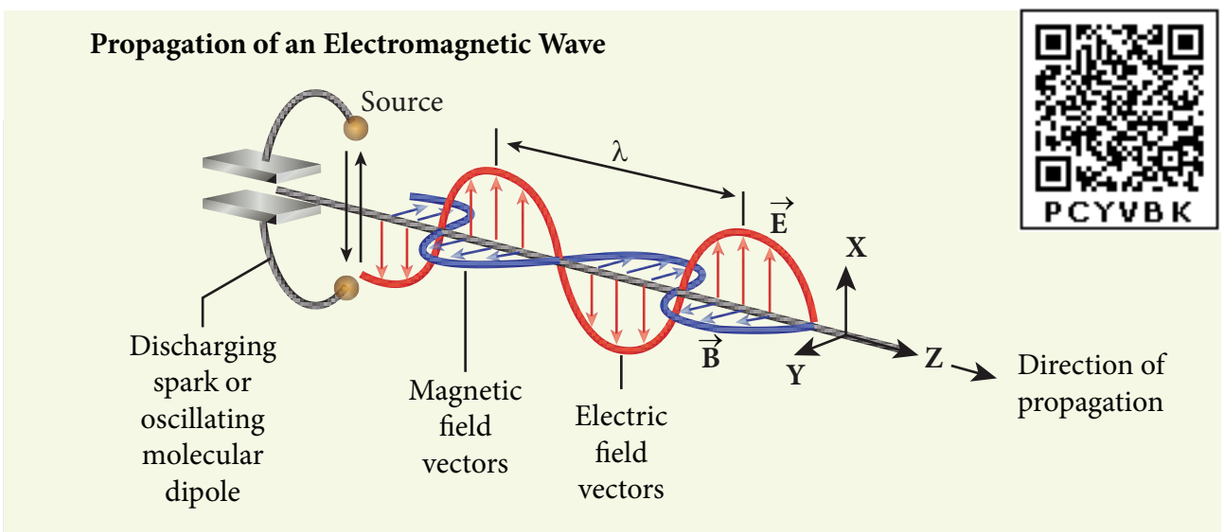


Figure 5.9 Oscillating charges - sources of electromagnetic waves

UNIT-5(XII-Physics_Vol-1).indd 274 14-02-2020 20:11:23